CS 182 - HW 1 - Jie Wen

1. Why Learning Rates Cannot be Too Big

(a) For what values of learning rate $\eta > 0$ is the recurrence (3) stable?

For the recurrence relation $w_{t+1} = (1 - 2\eta\sigma^2)w_t + 2\eta\sigma y$ to be stable, the absolute value of the coefficient of w_t must be less than 1:
$|1 - 2\eta\sigma^2| < 1$

Solve

$$|1 - 2\eta\sigma^2| < 1 \quad \Longrightarrow \quad 0 < \eta < \frac{1}{\sigma^2}.$$

So the learning rate must satisfy $0 < \eta < \frac{1}{\sigma^2}$.

(b) If η is below that upper bound, how fast does w_t converge to its final solution $w^* = \frac{y}{\sigma}$?

The error at iteration t is $e_t = w_t - w^*$. The error evolves as $e_{t+1} = (1 - 2\eta\sigma^2)e_t \implies e_t = (1 - 2\eta\sigma^2)^t e_0$.
To find the number of iterations t to get within a factor $(1 - \epsilon)$ of the solution, which is equivalent to the error should be less than a fraction ϵ of the initial error.

$$|e_t| \leq \epsilon |e_0|$$
$$|1 - 2\eta\sigma^2|^t \leq \epsilon$$
$$t \ln(|1 - 2\eta\sigma^2|) \leq \ln(\epsilon)$$
$$t \geq \frac{\ln(\epsilon)}{\ln(|1 - 2\eta\sigma^2|)}$$

(c) For what learning rates $\eta > 0$ will we converge? Which of the two σ_i is limiting our learning rate?

The updates are decoupled:

$$w[1]_{t+1} = (1 - 2\eta\sigma_l^2)w[1]_t + 2\eta\sigma_l y[1]$$
$$w[2]_{t+1} = (1 - 2\eta\sigma_s^2)w[2]_t + 2\eta\sigma_s y[2]$$

Stability requires:

$$|1 - 2\eta\sigma_l^2| < 1 \quad \text{and} \quad |1 - 2\eta\sigma_s^2| < 1$$

Since $\sigma_l \gg \sigma_s$, the stricter condition is:

$$\eta < \frac{1}{\sigma_l^2}$$

Therefore, the range of learning rates for convergence is $0 < \eta < \frac{1}{\sigma_l^2}$. The larger singular value, σ_l, is limiting the learning rate.

(d) For the previous problem, depending on η, σ_l, σ_s, which of the two dimensions is converging faster and which is converging slower?

The error decay factors are:

$$r_l = |1 - 2\eta\sigma_l^2|, \quad r_s = |1 - 2\eta\sigma_s^2|$$

Since $\sigma_l \gg \sigma_s$, for $\eta < \frac{1}{\sigma_l^2}$, $2\eta\sigma_l^2$ will be larger than $2\eta\sigma_s^2$.

$$\text{Smaller convergence factor} \implies \text{Faster convergence} \implies \begin{cases} \text{Dimension for } \sigma_l & \text{converges faster} \\ \text{Dimension for } \sigma_s & \text{converges slower} \end{cases}$$

(e) For what value of η will we get the fastest overall convergence to the solution?

The overall convergence is limited by the slowest converging dimension. To achieve the fastest overall convergence, choose η to make the convergence rates of the two dimensions equal.

$$|1 - 2\eta\sigma_l^2| = |1 - 2\eta\sigma_s^2|$$
$$-(1 - 2\eta\sigma_l^2) = 1 - 2\eta\sigma_s^2$$
$$\eta = \frac{1}{\sigma_l^2 + \sigma_s^2}$$

(f) Comment on what would happen if we had more parallel problems with σ_i that all were in between σ_l and σ_s? Would they influence the choice of possible learning rates or the learning rate with the fastest convergence?

The range of possible learning rates is always determined by the largest singular value, so it would still be $0 < \eta < \frac{1}{\sigma_l^2}$. The learning rate for the fastest overall convergence is determined by the largest and smallest singular values. Therefore, the intermediate singular values would not influence the choice of the optimal learning rate, which would remain $\eta = \frac{1}{\sigma_l^2+\sigma_s^2}$.

(g) Using what you know about the SVD, how is the simple scalar and parallel scalar problem analysis above relevant to solving general least-squares problems of the form $Xw \approx y$ using gradient descent?

Write the SVD $X = U\Sigma V^\top$. Gradient descent on $\|Xw - y\|^2$ updates

$$w_{t+1} = w_t - 2\eta X^\top (Xw_t - y) = (I - 2\eta X^T X)w_t + 2\eta X^T y$$

Change coordinates $z = V^\top w$ and $\tilde{y} = U^\top y$. In the V-basis the coordinates decouple:

$$z_{t+1}^{(i)} = (1 - 2\eta\sigma_i^2) z_t^{(i)} + 2\eta\sigma_i \tilde{y}_i,$$

where σ_i are the singular values (diagonal entries of Σ). This is **exactly** the parallel scalar problem analyzed above.

Consequences:

- The step-size must satisfy $0 < \eta < 1/\sigma_{\max}^2$ (where $\sigma_{\max}$ is the largest singular value of X).
- Convergence speed is dominated by the smallest singular values: the condition number $\kappa = \sigma_{\max}^2/\sigma_{\min}^2$ determines how many iterations are needed (large $\kappa \to$ slow).
- The best minimax η balances extremes (as in part (e)); interior singular values do not change that minimax choice.

So the scalar and parallel analyses are the exact local picture of gradient descent for least squares after rotating into the SVD basis: the singular values σ_i play the same role as the scalar σ.

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2. Stochastic Gradient Descent (when it is possible to interpolate)

(a) Preliminaries and Change of Coordinates

The system is

$$Xw = y,$$

where $X \in \mathbb{R}^{n\times d}$ has full row rank (so $n \le d$), and $y \in \mathbb{R}^n$. The min-norm solution is

$$w^* = X^T(XX^T)^{-1}y.$$

Define the shifted variable

$$w' = w - w^*.$$

Then

$$Xw' = X(w - w^*) = Xw - Xw^* = y - y = 0.$$

The initial condition is $w_0 = 0$, so

$$w_0' = -w^* = -X^T(XX^T)^{-1}y.$$

(b) SVD Transformation

Let the SVD of X be

$$X = U\Sigma V^T,$$

where $U \in \mathbb{R}^{n\times n}$ and $V \in \mathbb{R}^{d\times d}$ are orthogonal, and $\Sigma = [D \quad 0] \in \mathbb{R}^{n\times d}$ with D diagonal and positive. Define

$$w'' = V^T w'.$$

Then

$$Xw' = U\Sigma V^T w' = U\Sigma w'' = 0 \Rightarrow \Sigma w'' = 0.$$

Since $\Sigma = [D \quad 0]$, this implies the first n components of w'' must be zero. The initial condition becomes

$$w_0'' = V^T w_0' = -V^T w^* = -[D^{-1}U^T y;\ 0],$$

so the last $d - n$ components are zero.

(c) Reduced Square System

Because the last $d-n$ coordinates of w'' are always irrelevant to the equation (they multiply the zero block in $[\tilde{X}\ 0]$), the only degrees of freedom that matter are the first n coordinates of w''. Denote the first n coordinates by $\tilde{w} \in \mathbb{R}^n$. Then the system reduces to the square $n \times n$ system

$$\tilde{X}\,\tilde{w} = 0,$$

which is (8). (Here $\tilde{X} = \mathrm{diag}(\sigma_1, \ldots, \sigma_n)$ is invertible because $\sigma_i > 0$.)

Each row of $\tilde{X}\tilde{w} = 0$ is just an orthonormal change of coordinates of the corresponding original row equation in $Xw = y$ (applied the left multiplication by U^T and right multiplication by V^T to move to SVD coordinates). So the i-th equation of (8) corresponds, via these orthonormal transforms, to the i-th equation of the original system — i.e. it is the same linear constraint expressed in the new coordinates.

(d) SGD Update in Transformed Coordinates

The loss on the i-th example is

$$L_i(w) = (y_i - x_i^T w)^2.$$

In shifted coordinates $w' = w - w^*$, since $x_i^T w^* = y_i$:

$$L_i(w) = (x_i^T(w^* - w))^2 = (x_i^T(-w'))^2 = (x_i^T w')^2.$$

The gradient is

$$\nabla L_i(w) \;=\; -2x_i(y_i - x_i^T w) \;=\; 2x_i x_i^T w'.$$

The SGD update is

$$w'_{t+1} = w'_t - 2\eta(x_i^T w'_t)x_i.$$

Now move to SVD coordinates $w'' = V^T w'$ and $x_i'' := V^T x_i$. Because V is orthonormal, gradients transform as $V^T \nabla_w L_i(w) = \nabla_{w''} L_i''(w'')$. The SGD update in the $\tilde{w}$ / $\tilde{X}$ coordinates becomes

$$w''_{t+1} = w''_t - \eta V^T \nabla L_{I_t}(w_t) = w''_t - \eta \cdot 2\, x''_{I_t} x''^{\,T}_{I_t} w''_t.$$

Restricting to the first n coordinates (the only nonzero ones in w''_0) and writing $\tilde{x}_{I_t}$ for the corresponding n-vector (the i-th row/column in the diagonal case becomes $\tilde{x}_i = \sigma_i e_i$):

$$\tilde{w}_{t+1} = \tilde{w}_t - 2\eta\, \tilde{x}_{I_t} \tilde{x}_{I_t}^T \tilde{w}_t,$$

which is (10). So SGD steps are identical in the transformed coordinates (because U, V are orthonormal).

(e) Stochastic Lyapunov Function and Convergence

If we have a Lyapunov function $L(\tilde{w}_t) \geq 0$ satisfying

$$\mathbb{E}[L(\tilde{w}_{t+1}) \mid \tilde{w}_t] \leq (1-\rho)\, L(\tilde{w}_t) \quad \text{with } 0 < \rho < 1,$$

then by taking expectations and iterating:

$$\mathbb{E}[L(\tilde{w}_t)] \leq (1-\rho)^t L(\tilde{w}_0).$$

By Markov's inequality,

$$\mathbb{P}\big(L(\tilde{w}_t) \geq \varepsilon\big) \leq \frac{\mathbb{E}[L(\tilde{w}_t)]}{\varepsilon} \leq \frac{(1-\rho)^t L(\tilde{w}_0)}{\varepsilon}.$$

Thus choose T large enough that

$$\frac{(1-\rho)^T L(\tilde{w}_0)}{\varepsilon} \leq \delta,$$

i.e.

$$(1-\rho)^T \leq \frac{\delta\varepsilon}{L(\tilde{w}_0)}.$$

Such a T exists (take logarithms). Then

$$\mathbb{P}\big(L(\tilde{w}_T) < \varepsilon\big) \geq 1 - \delta,$$

which is (12).

(f) Lyapunov Function Candidate

$L(\tilde{w}) = \tilde{w}^T \tilde{X}^T \tilde{X} \tilde{w} = \|\tilde{X}\tilde{w}\|^2 \geq 0$, with equality iff $\tilde{X}\tilde{w} = 0$. Since $\tilde{X}$ is $n \times n$ full rank (invertible), this holds iff $\tilde{w} = 0$.

(g) Expansion of Lyapunov Function

Let $\Delta := \tilde{w}_{t+1} - \tilde{w}_t$. Then

$$\begin{aligned} L(\tilde{w}_{t+1}) &= (\tilde{w}_t + \Delta)^T \tilde{X}^T \tilde{X} (\tilde{w}_t + \Delta) \\ &= \tilde{w}_t^T \tilde{X}^T \tilde{X} \tilde{w}_t \;+\; 2\tilde{w}_t^T \tilde{X}^T \tilde{X} \Delta \;+\; \Delta^T \tilde{X}^T \tilde{X} \Delta. \end{aligned}$$

So taking $L(\tilde{w}_t) = \tilde{w}_t^T \tilde{X}^T \tilde{X} \tilde{w}_t$, define

$$A = 2\tilde{w}_t^T \tilde{X}^T \tilde{X} \Delta, \quad B = \Delta^T \tilde{X}^T \tilde{X} \Delta,$$

giving $L(\tilde{w}_{t+1}) = L(\tilde{w}_t) + A + B$ as requested.

(h) Expectation of Linear Term A

Recall the single-sample SGD update (10):

$$\Delta = \tilde{w}_{t+1} - \tilde{w}_t = -2\eta \, \tilde{x}_{I_t} \tilde{x}_{I_t}^T \tilde{w}_t.$$

Then

$$A = 2\tilde{w}_t^T \tilde{X}^T \tilde{X} \Delta = -4\eta \, \tilde{w}_t^T \tilde{X}^T \tilde{X} \big(\tilde{x}_{I_t} \tilde{x}_{I_t}^T\big) \tilde{w}_t.$$

Take conditional expectation over the uniform random index I_t:

$$\mathbb{E}[A \mid \tilde{w}_t] = -\frac{4\eta}{n} \sum_{i=1}^{n} \tilde{w}_t^T \tilde{X}^T \tilde{X} \, \tilde{x}_i \tilde{x}_i^T \tilde{w}_t = -\frac{4\eta}{n} \, \tilde{w}_t^T (\tilde{X}^T \tilde{X})(\tilde{X}^T \tilde{X}) \tilde{w}_t.$$

That is

$$\mathbb{E}[A \mid \tilde{w}_t] = -\frac{4\eta}{n} \, \tilde{w}_t^T (\tilde{X}^T \tilde{X})^2 \tilde{w}_t.$$

Let $\lambda_{\min}$ denote the smallest eigenvalue of $\tilde{X}^T \tilde{X}$. (Since $\tilde{X}$ is invertible, $\lambda_{\min} > 0$.) Then

$$(\tilde{X}^T \tilde{X})^2 \succeq \lambda_{\min} \, (\tilde{X}^T \tilde{X}),$$

so

$$\tilde{w}_t^T (\tilde{X}^T \tilde{X})^2 \tilde{w}_t \geq \lambda_{\min} \, \tilde{w}_t^T (\tilde{X}^T \tilde{X}) \tilde{w}_t = \lambda_{\min} \, L(\tilde{w}_t).$$

Therefore

$$\mathbb{E}[A \mid \tilde{w}_t] \leq -\frac{4\eta}{n} \, \lambda_{\min} \, L(\tilde{w}_t).$$

Set $c_1 := \dfrac{4\lambda_{\min}}{n} > 0$ and obtain (15):

$$\mathbb{E}[A \mid \tilde{w}_t] \leq -c_1 \eta \, L(\tilde{w}_t).$$

(Interpretation: the linear term yields a descent proportional to $\eta L(\tilde{w}_t)$.)

(i) Expectation of Quadratic Term B

Using $\Delta = -2\eta \tilde{x}_{I_t} \tilde{x}_{I_t}^T \tilde{w}_t$,

$$\begin{aligned} B = \Delta^T \tilde{X}^T \tilde{X} \Delta &= 4\eta^2 \, (\tilde{w}_t^T \tilde{x}_{I_t}) \, (\tilde{x}_{I_t}^T \tilde{X}^T \tilde{X} \tilde{x}_{I_t}) \, (\tilde{x}_{I_t}^T \tilde{w}_t) \\ &= 4\eta^2 \, (\tilde{x}_{I_t}^T \tilde{w}_t)^2 \, \big(\tilde{x}_{I_t}^T \tilde{X}^T \tilde{X} \tilde{x}_{I_t}\big). \end{aligned}$$

Introduce:

- $\lambda_{\max}$ = largest eigenvalue of $\tilde{X}^T \tilde{X}$,
- $\beta := \max_i \|\tilde{x}_i\|_2^2$ (rows' squared norm maximum).

Then $\tilde{x}_{I_t}^T \tilde{X}^T \tilde{X} \tilde{x}_{I_t} \leq \lambda_{\max} \|\tilde{x}_{I_t}\|_2^2 \leq \lambda_{\max} \beta$.
Also $(\tilde{x}_{I_t}^T \tilde{w}_t)^2 \leq \|\tilde{x}_{I_t}\|_2^2 \|\tilde{w}_t\|_2^2 \leq \beta \|\tilde{w}_t\|_2^2$.

Combine:

$$B \le 4\eta^2 \left(\beta \|\tilde{w}_t\|_2^2\right)\left(\lambda_{\max}\beta\right) = 4\eta^2 \lambda_{\max}\beta^2 \, \|\tilde{w}_t\|_2^2.$$

Relate $\|\tilde{w}_t\|_2^2$ to $L(\tilde{w}_t)$: since $\tilde{X}^T\tilde{X} \succeq \lambda_{\min} I$,

$$L(\tilde{w}_t) = \tilde{w}_t^T \tilde{X}^T \tilde{X} \tilde{w}_t \ge \lambda_{\min} \|\tilde{w}_t\|_2^2,$$

so $\|\tilde{w}_t\|_2^2 \le \frac{1}{\lambda_{\min}} L(\tilde{w}_t)$. Thus

$$B \le 4\eta^2 \lambda_{\max}\beta^2 \cdot \frac{1}{\lambda_{\min}} L(\tilde{w}_t) = \underbrace{\frac{4\lambda_{\max}\beta^2}{\lambda_{\min}}}_{=:c_2} \eta^2 \, L(\tilde{w}_t).$$

Taking conditional expectation (the bound did not depend on the random index beyond β):

$$\mathbb{E}[B \mid \tilde{w}_t] \le c_2 \eta^2 \, L(\tilde{w}_t),$$

which is (16). (Here $c_2 > 0$ depends on the problem via $\lambda_{\max}, \lambda_{\min}, \beta$.)

(j) Combining the Bounds

From (g)–(i),

$$\mathbb{E}[L(\tilde{w}_{t+1}) \mid \tilde{w}_t] = L(\tilde{w}_t) + \mathbb{E}[A \mid \tilde{w}_t] + \mathbb{E}[B \mid \tilde{w}_t] \le \left(1 - c_1\eta + c_2\eta^2\right) L(\tilde{w}_t).$$

If picking $\eta > 0$ small enough that

$$c_1\eta > c_2\eta^2 \quad \Longleftrightarrow \quad 0 < \eta < \frac{c_1}{c_2},$$

then $1 - c_1\eta + c_2\eta^2 < 1$. Thus for such small constant η, it's ok to obtain a uniform multiplicative contraction in expectation:

$$\mathbb{E}[L(\tilde{w}_{t+1}) \mid \tilde{w}_t] \le (1 - \rho)\, L(\tilde{w}_t) \quad \text{with } \rho := c_1\eta - c_2\eta^2 \in (0, 1).$$

By part (e) this yields the high-probability guarantee that $L(\tilde{w}_t)$ (and hence $\tilde{w}_t$) becomes arbitrarily small with high probability in $O(\log(1/\delta)/\rho)$ iterations.

(k) Observations from Demo Notebook

Based on the Jupyter Notebook, the **Feature Augmented** method for ridge regression converges significantly faster than applying SGD directly to the **Original Ridge** objective.

Here is the explanation:

1. **Exponential Convergence for Feature Augmentation:** The feature augmentation trick transforms ridge regression into an over-parameterized, standard least-squares problem where a zero-loss solution exists (it can interpolate the data). As proven in parts (a)-(j) of this problem, SGD with a constant step size achieves rapid, **exponential convergence** in this interpolating regime. This is visible as a straight line on the log-loss plot.
2. **Slower Convergence for Naive Ridge SGD:** Applying SGD to the standard ridge objective is a non-interpolating problem, as the optimal loss is non-zero. With a constant step size, SGD does not converge precisely to the minimizer but instead fluctuates within a "noise ball" around it. This results in much slower, sub-linear convergence and requires a diminishing step size to converge accurately.

The core lesson is: Mathematically equivalent formulations can have vastly different algorithmic properties. The feature augmentation trick creates a structure that is far more suitable for fast convergence with constant-step-size SGD.

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3. Accelerating Gradient Descent with Momentum

(a) Reparameterization of the Momentum Update

The momentum update equations are:

$$w_{t+1} = w_t - \eta z_{t+1}, \quad z_{t+1} = (1-\beta) z_t + \beta g_t,$$

where $g_t = \nabla L(w_t) = 2X^T(Xw_t - y)$. Using the SVD $X = U\Sigma V^T$, define the shifted variables:

$$x_t = V^T(w_t - w^*), \quad a_t = V^T z_t,$$

where $w^* = (X^TX)^{-1}X^Ty$ is the minimizer. Then, for each component i, the recurrence becomes:

$$a_{t+1}[i] = (1-\beta) a_t[i] + 2\beta\sigma_i^2 x_t[i],$$
$$x_{t+1}[i] = x_t[i] - \eta a_{t+1}[i].$$

(b) Matrix Form of the Recurrence

Write the recurrence in vector form

$$\begin{bmatrix} a_{t+1}[i] \\ x_{t+1}[i] \end{bmatrix} = R_i \begin{bmatrix} a_t[i] \\ x_t[i] \end{bmatrix}.$$

From the equations in **(a)**:

$$a_{t+1} = (1-\beta)a_t + 2\beta\sigma_i^2 x_t, \qquad x_{t+1} = -\eta(1-\beta)a_t + (1-2\eta\beta\sigma_i^2)x_t.$$

So

$$R_i = \begin{bmatrix} 1-\beta & 2\beta\sigma_i^2 \\ -\eta(1-\beta) & 1-2\eta\beta\sigma_i^2 \end{bmatrix}.$$

(c) Eigenvalues of R_i

The eigenvalues of R_i are given by:

$$\lambda = \frac{(2-\beta-2\eta\beta\sigma_i^2) \pm \sqrt{(2-\beta-2\eta\beta\sigma_i^2)^2 - 4(1-\beta)}}{2}.$$

- **Real eigenvalues** when $(2-\beta-2\eta\beta\sigma_i^2)^2 \geq 4(1-\beta)$,
- **Repeated eigenvalues** when equality holds,
- **Complex eigenvalues** when $(2-\beta-2\eta\beta\sigma_i^2)^2 < 4(1-\beta)$.

(d) Repeated Eigenvalues

For repeated eigenvalues, the condition is:

$$(2-\beta-2\eta\beta\sigma_i^2)^2 = 4(1-\beta).$$

Solving for η, the highest learning rate that results in repeated eigenvalues is:

$$\eta = \frac{2-\beta+2\sqrt{1-\beta}}{2\beta\sigma_i^2}.$$

This is stable since $\sqrt{1-\beta} < 1$ for $\beta > 0$.

(e) Real Eigenvalues

For real eigenvalues, stability requires:

$$|2-\beta-2\eta\beta\sigma_i^2| < 2-\beta,$$

which simplifies to:

$$\eta < \frac{2-\beta}{\beta\sigma_i^2}.$$

The range of η for real and stable eigenvalues is:

$$0 < \eta \leq \frac{2-\beta-2\sqrt{1-\beta}}{2\beta\sigma_i^2} \quad \text{or} \quad \frac{2-\beta+2\sqrt{1-\beta}}{2\beta\sigma_i^2} \leq \eta < \frac{2-\beta}{\beta\sigma_i^2}.$$

(f) Complex Eigenvalues

For complex eigenvalues, the magnitude is $\sqrt{1-\beta} < 1$ for $\beta > 0$, so stability is automatic. The highest learning rate for complex eigenvalues is approached from below:

$$\eta \to \frac{2-\beta+2\sqrt{1-\beta}}{2\beta\sigma_i^2}.$$

(g) Application to Given Parameters

Given $\beta = 0.1$, $\sigma_{\max} = 5$, $\sigma_{\min} = 0.05$, the optimal learning rate is close to the stability limit:

$$\eta \approx \frac{2-\beta}{\beta\sigma_{\max}^2} = \frac{1.9}{0.1 \cdot 25} = 0.76.$$

With this choice, the spectral radius is approximately 0.9996. The number of iterations required to reduce the error to 0.1% of the initial value is:

$$t = \frac{\log(0.001)}{\log(0.9996)} \approx 17316.$$

For ordinary gradient descent with optimal learning rate $\eta^* = \frac{2}{\sigma_{\max}^2 + \sigma_{\min}^2} \approx 0.08$, the spectral radius is:

$$\rho = \frac{(\sigma_{\max}/\sigma_{\min})^2 - 1}{(\sigma_{\max}/\sigma_{\min})^2 + 1} = \frac{10000 - 1}{10000 + 1} \approx 0.9998,$$

and the number of iterations required is:

$$t = \frac{\log(0.001)}{\log(0.9998)} \approx 34539.$$

Thus, momentum with $\beta = 0.1$ roughly halves the number of iterations.

(h) Notebook-Based Questions

(i) How does σ_i influence the gradients and parameter updates?

The eigenvalues σ_i determine the curvature of the loss landscape. Directions with large σ_i have high curvature, leading to large gradients and rapid parameter changes. Directions with small σ_i have low curvature, resulting in small gradients and slow parameter changes.

(ii) Which converges faster, gradient descent or gradient descent with momentum? Why?

Gradient descent with momentum converges faster because it smooths the gradients and allows larger effective step sizes, particularly in directions with small eigenvalues, accelerating convergence.

(iii) Can the learning rate be changed to further accelerate convergence?

Yes, increasing the learning rate for momentum can further accelerate convergence, but care must be taken to remain within stability limits.

(i) Comparing gradient descent and gradient descent with momentum, which one converges faster for this task? Why?

Gradient descent with momentum converges faster for this task because momentum accumulates gradients, allowing larger effective steps in low-curvature directions (small σ_i), reducing the impact of ill-conditioning and accelerating overall progress.

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4. Optimizers

(a) Completing the Adam Pseudocode

(A): $\beta_1 m_{t-1} + (1 - \beta_1) g_t$
(B): $\beta_2 v_{t-1} + (1 - \beta_2) g_t^2$

(b) Equivalence Between Weight Decay and L2 Regularization

Compare two update rules:

1. **SGD on L2-regularized loss**:

   $$f_t^{\text{reg}}(\theta) = f_t(\theta) + \frac{\lambda}{2}\|\theta\|_2^2$$

   The gradient is:

   $$\nabla f_t^{\text{reg}}(\theta) = \nabla f_t(\theta) + \lambda\theta$$

   The update rule becomes:

   $$\theta_{t+1} = \theta_t - \eta \nabla f_t^{\text{reg}}(\theta_t) = \theta_t - \eta \nabla f_t(\theta_t) - \eta\lambda\theta_t$$

2. **SGD with weight decay**:

   $$\theta_{t+1} = (1 - \gamma)\theta_t - \eta \nabla f_t(\theta_t)$$

To make these equivalent, equate the coefficients of θ_t:

$$\theta_t - \eta\lambda\theta_t = (1 - \gamma)\theta_t$$

Simplifying:

$$1 - \eta\lambda = 1 - \gamma \quad \Rightarrow \quad \gamma = \eta\lambda$$

Thus, setting $\gamma = \eta\lambda$ makes the two update rules equivalent.

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5. Regularization and Instance Noise

(a) Equivalence to Regularized Least Squares

Let $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{N}$, where $\mathbf{X}$ is the fixed data matrix and $\mathbf{N}$ is a noise matrix with rows $\mathbf{N}_i^\top \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$. Then:

$$\|\tilde{\mathbf{X}}\mathbf{w} - \mathbf{y}\|^2 = (\tilde{\mathbf{X}}\mathbf{w} - \mathbf{y})^\top (\tilde{\mathbf{X}}\mathbf{w} - \mathbf{y}) = \mathbf{w}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{w} - 2\mathbf{y}^\top \tilde{\mathbf{X}} \mathbf{w} + \mathbf{y}^\top \mathbf{y}$$

Taking expectation:

$$\mathbb{E}[\|\tilde{\mathbf{X}}\mathbf{w} - \mathbf{y}\|^2] = \mathbf{w}^\top \mathbb{E}[\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}] \mathbf{w} - 2\mathbf{y}^\top \mathbb{E}[\tilde{\mathbf{X}}] \mathbf{w} + \mathbf{y}^\top \mathbf{y}$$

Since $\mathbb{E}[\mathbf{N}] = 0$:

$$\mathbb{E}[\tilde{\mathbf{X}}] = \mathbf{X}, \quad \mathbb{E}[\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}] = \mathbf{X}^\top \mathbf{X} + \mathbb{E}[\mathbf{N}^\top \mathbf{N}]$$

Each row of $\mathbf{N}$ is i.i.d. $\mathcal{N}(0, \sigma^2 \mathbf{I}_n)$, so:

$$\mathbb{E}[\mathbf{N}^\top \mathbf{N}] = m\sigma^2 \mathbf{I}_n$$

Thus:

$$\mathbb{E}[\|\tilde{\mathbf{X}}\mathbf{w} - \mathbf{y}\|^2] = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + m\sigma^2 \|\mathbf{w}\|^2$$

Dividing by m:

$$\frac{1}{m} \mathbb{E}[\|\tilde{\mathbf{X}}\mathbf{w} - \mathbf{y}\|^2] = \frac{1}{m} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \sigma^2 \|\mathbf{w}\|^2$$

So the problem is equivalent to:

$$\min_{\mathbf{w}} \left(\frac{1}{m} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|^2 \right), \quad \text{with } \lambda = \sigma^2$$

(b) Gradient Descent Recurrence

Take conditional expectation given w_t (so only randomness is $\tilde{X}_t$):

$$\mathbb{E}[w_{t+1} \mid w_t] = w_t - \eta \Big(\mathbb{E}[\tilde{X}_t^2]\, w_t - y\, \mathbb{E}[\tilde{X}_t] \Big).$$

Compute moments: $\mathbb{E}[\tilde{X}_t] = x$, $\mathbb{E}[\tilde{X}_t^2] = x^2 + \sigma^2$. Thus

$$\mathbb{E}[w_{t+1} \mid w_t] = w_t - \eta \big((x^2 + \sigma^2) w_t - yx \big).$$

Now take expectation over w_t:

$$\mathbb{E}[w_{t+1}] = \big(1 - \eta(x^2 + \sigma^2)\big)\, \mathbb{E}[w_t] + \eta\, yx.$$

(c) Convergence Condition

The recurrence:

$$\mathbb{E}[w_{t+1}] = A \cdot \mathbb{E}[w_t] + B, \quad \text{where } A = 1 - \eta(x^2 + \sigma^2), \quad B = \eta xy$$

For convergence, require $|A| < 1$:

$$-1 < 1 - \eta(x^2 + \sigma^2) < 1$$

From the right inequality:

$$1 - \eta(x^2 + \sigma^2) < 1 \Rightarrow \eta > 0$$

From the left inequality:

$$1 - \eta(x^2 + \sigma^2) > -1 \Rightarrow \eta(x^2 + \sigma^2) < 2$$

Thus:

$$0 < \eta < \frac{2}{x^2 + \sigma^2}$$

(d) Limit of Expected Weight

At convergence, $\mathbb{E}[w_{t+1}] = \mathbb{E}[w_t] = w^*$. Solving:

$$w^* = (1 - \eta(x^2 + \sigma^2))w^* + \eta xy$$
$$\eta(x^2 + \sigma^2)w^* = \eta xy \Rightarrow w^* = \frac{xy}{x^2 + \sigma^2}$$

Without noise ($\sigma = 0$), the optimal weight would be:

$$w^* = \frac{y}{x}$$

The noise acts as regularization, shrinking the weight estimate.

6: General Case Tikhonov Regularization

(a) Expand, take gradient, solve.

Write the objective

$$f(\mathbf{x}) = \|W_1(A\mathbf{x} - \mathbf{b})\|_2^2 + \|W_2(\mathbf{x} - \mathbf{c})\|_2^2.$$

As matrix products,

$$f(\mathbf{x}) = (A\mathbf{x} - \mathbf{b})^T W_1^T W_1 (A\mathbf{x} - \mathbf{b}) + (\mathbf{x} - \mathbf{c})^T W_2^T W_2 (\mathbf{x} - \mathbf{c}).$$

Differentiate (use $\nabla_x (x^T M x) = (M + M^T)x$ and M here will be symmetric when $M = W_i^T W_i$):

$$\nabla f(\mathbf{x}) = 2A^T W_1^T W_1 (A\mathbf{x} - \mathbf{b}) + 2W_2^T W_2 (\mathbf{x} - \mathbf{c}).$$

Set gradient to zero:

$$A^T W_1^T W_1 (A\mathbf{x} - \mathbf{b}) + W_2^T W_2 (\mathbf{x} - \mathbf{c}) = 0.$$

Collect terms in $\mathbf{x}$:

$$\left(A^T W_1^T W_1 A + W_2^T W_2\right)\mathbf{x} = A^T W_1^T W_1 \mathbf{b} + W_2^T W_2 \mathbf{c}.$$

Thus (when the left matrix is invertible) the minimizer is

$$\mathbf{x}^* = \left(A^T W_1^T W_1 A + W_2^T W_2\right)^{-1} \left(A^T W_1^T W_1 \mathbf{b} + W_2^T W_2 \mathbf{c}\right).$$

(b) Recast as a single least-squares and recover the same solution.

Define

$$\mathbf{C} = \begin{bmatrix} W_1 A \\ W_2 \end{bmatrix}, \qquad \mathbf{d} = \begin{bmatrix} W_1 \mathbf{b} \\ W_2 \mathbf{c} \end{bmatrix}.$$

Then

$$\|\mathbf{C}\mathbf{x} - \mathbf{d}\|_2^2 = \|W_1(A\mathbf{x} - \mathbf{b})\|_2^2 + \|W_2(\mathbf{x} - \mathbf{c})\|_2^2,$$

so the problems are identical.

The normal equations for this OLS problem are

$$(C^T C)\mathbf{x} = C^T \mathbf{d}.$$

Compute

$$C^T C = (W_1 A)^T (W_1 A) + W_2^T W_2 = A^T W_1^T W_1 A + W_2^T W_2,$$
$$C^T d = (W_1 A)^T (W_1 b) + W_2^T (W_2 c) = A^T W_1^T W_1 b + W_2^T W_2 c.$$

So the OLS solution

$$\mathbf{x}^* = (C^T C)^{-1} C^T \mathbf{d}$$

is exactly the same expression as in (a). This confirms agreement.

(c) Reduce to ridge regression.

Ridge is

$$\min_x \|Ax - b\|_2^2 + \lambda \|x\|_2^2,$$

whose solution is $x^* = (A^T A + \lambda I)^{-1} A^T b$.

Choose

$$W_1 = I, \qquad W_2 = \sqrt{\lambda}\, I, \qquad \mathbf{c} = \mathbf{0}.$$

Then $W_1^T W_1 = I$ and $W_2^T W_2 = \lambda I$, and the formula from (a) becomes

$$x^* = (A^T A + \lambda I)^{-1}(A^T b + \lambda \cdot 0) = (A^T A + \lambda I)^{-1} A^T b,$$

which is exactly ridge regression.

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7. An Alternate MAP Interpretation of Ridge Regression

Since $\mathbf{W}$ and $\mathbf{Y}$ are jointly normal (as linear transformations of Gaussian vectors), the conditional distribution $\mathbf{W} \mid \mathbf{Y} = \mathbf{y}$ is also normal.

For a normal distribution, the **mode** (MAP) equals the **mean**. Therefore:

$$\hat{\mathbf{w}}_{\text{MAP}} = \mathbb{E}[\mathbf{W} \mid \mathbf{Y} = \mathbf{y}]$$

For zero-mean jointly normal vectors, the conditional expectation is:

$$\mathbb{E}[\mathbf{W} \mid \mathbf{Y} = \mathbf{y}] = \Sigma_{WY} \Sigma_{YY}^{-1} \mathbf{y} \tag{32}$$

where:

- $\Sigma_{YY} = \mathbb{E}[\mathbf{Y}\mathbf{Y}^T]$
- $\Sigma_{WY} = \mathbb{E}[\mathbf{W}\mathbf{Y}^T]$

Given:

$$\mathbf{Y} = X\mathbf{W} + \sqrt{\lambda}\mathbf{N}$$

with $\mathbf{W} \sim \mathcal{N}(0, I)$, $\mathbf{N} \sim \mathcal{N}(0, I)$, and $\mathbf{W} \perp \mathbf{N}$.

$$\begin{aligned}
\Sigma_{YY} &= \mathbb{E}[\mathbf{Y}\mathbf{Y}^T] \\
&= \mathbb{E}[(X\mathbf{W} + \sqrt{\lambda}\mathbf{N})(X\mathbf{W} + \sqrt{\lambda}\mathbf{N})^T] \\
&= X\mathbb{E}[\mathbf{W}\mathbf{W}^T]X^T + \sqrt{\lambda}X\mathbb{E}[\mathbf{W}\mathbf{N}^T] + \sqrt{\lambda}\mathbb{E}[\mathbf{N}\mathbf{W}^T]X^T + \lambda\mathbb{E}[\mathbf{N}\mathbf{N}^T] \\
&= XIX^T + 0 + 0 + \lambda I \\
&= XX^T + \lambda I
\end{aligned}$$

$$\begin{aligned}
\Sigma_{WY} &= \mathbb{E}[\mathbf{W}\mathbf{Y}^T] \\
&= \mathbb{E}[\mathbf{W}(X\mathbf{W} + \sqrt{\lambda}\mathbf{N})^T] \\
&= \mathbb{E}[\mathbf{W}\mathbf{W}^T]X^T + \sqrt{\lambda}\mathbb{E}[\mathbf{W}\mathbf{N}^T] \\
&= IX^T + 0 \\
&= X^T
\end{aligned}$$

From (32):

$$\mathbb{E}[\mathbf{W} \mid \mathbf{Y} = \mathbf{y}] = X^T(XX^T + \lambda I)^{-1}\mathbf{y} \tag{31}$$

But since the MAP estimate equals the conditional mean, The two forms of the Ridge Regression solution are equivalent:

$$(X^T X + \lambda I)^{-1} X^T \mathbf{y} = X^T (XX^T + \lambda I)^{-1} \mathbf{y}$$

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8. Homework Process and Study Group

(a)

- In Question 2, I asked Gemini: *"What is the purpose of the coordinate transformations in parts 2(a)-2(c)? Why does it simplify the problem?"*
  - The AI explained that the SVD diagonalizes the system. In the new coordinate system (w''), the problem decouples into independent scalar problems. This allows us to analyze a much simpler $n \times n$ system (Equation 8) instead of the original $d \times d$ system, without loss of generality.
- In Question 3, the eigenvalue analysis in parts (c)-(f) was complex. I used the AI to explore concepts: *"For the complex eigenvalue case, the magnitude is $\sqrt{1-\beta}$. What does this tell us about the relationship between the momentum parameter β and the stability of the algorithm?"* Discussing this showed me that a larger β (closer to 1)

leads to faster convergence in the oscillatory, complex-valued regime. This deeper conceptual understanding was far more valuable than just being given the stability conditions.

(b)

None

(c)

5h..